# Introduction to Economics

Class 9 — Price controls; international trade and tariffs

**Pavel Kocourek**

Universitat de Barcelona · Group A6

22 October 2026

# Plan for today

Warm-up
Catalonia caps rents: what will happen?
The lost triangle, computed
In pairs: taxes on menús
A rent cap: a shortage, now and over the years
Catalonia: what the data say
Break · Problem set 1 on the front desk
A price floor: the minimum wage
Open borders: who wins, who loses
In pairs: cava at home and abroad
A tariff: who gains, what is lost
EU car duties; why limit trade; in pairs
In one line; before Monday 26 October

## Warm-up: same goal, opposite advice

### Three economists all want young workers to earn more

A: raise the minimum wage.
B: don't: firms would hire fewer young people.

1. B's objection: values or a prediction?

2. C: "A few jobs lost is a fair price for higher pay." Values or a prediction?

In pairs, one minute; then hands.

## Rooms for students in Barcelona

1. A new campus brings 2,000 more students
2. Owners turn rooms into tourist lets
3. Because of 2, rents rise: does demand shift too?

In pairs, two minutes; then hands per item.

# 16 March 2024: Catalonia caps the rent on new contracts

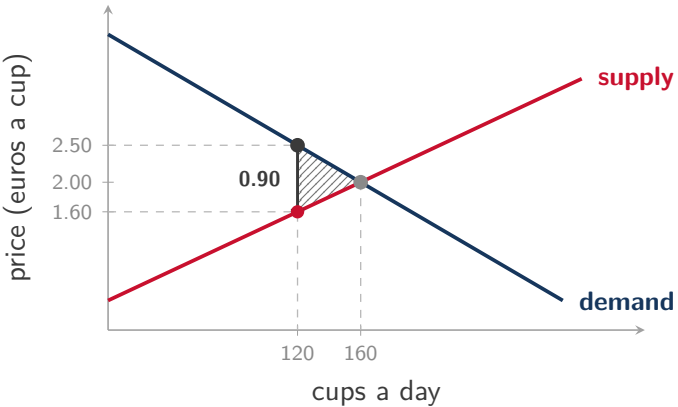



140 towns with a stressed rental market, Barcelona included

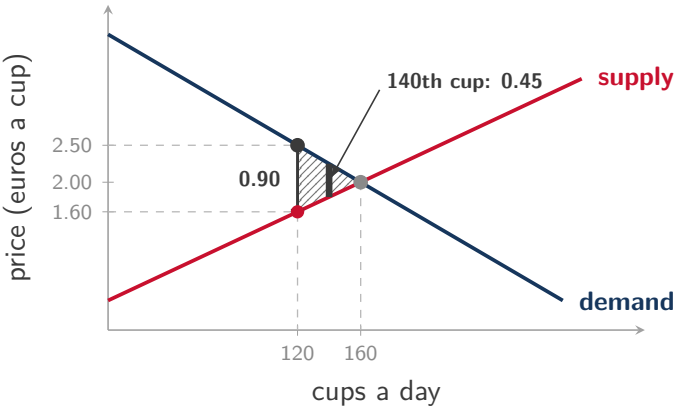
**What happens to rents —**
**and to how many flats get rented?**

Ley 12/2023 (Spain's housing law); Resolució TER/800/2024 (DOGC, 14 March 2024), in force 16 March 2024.

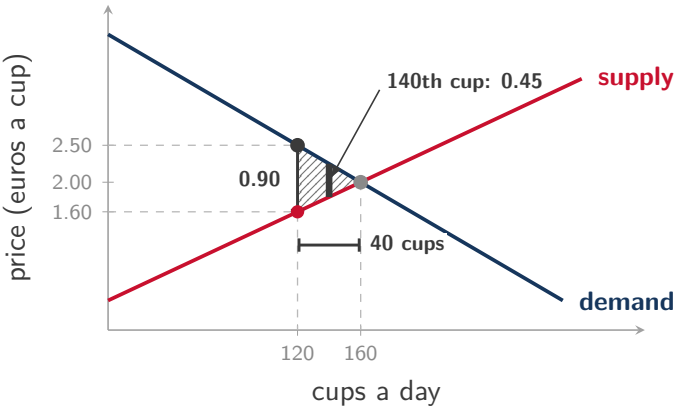
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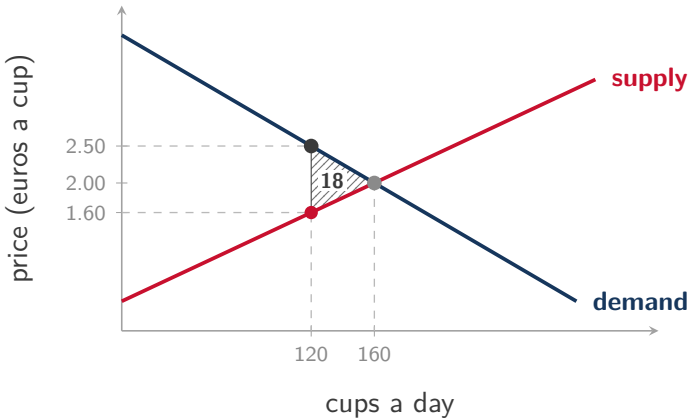


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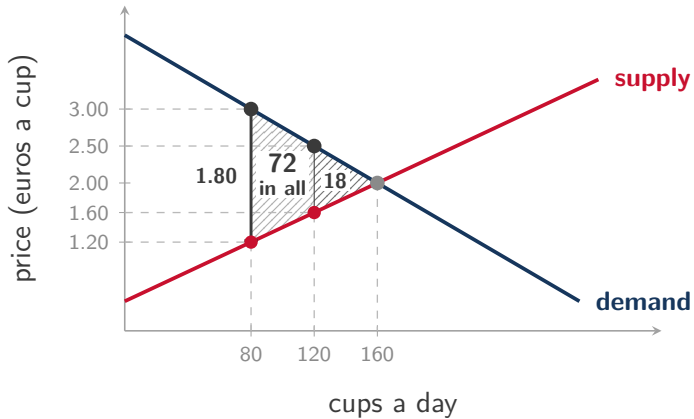



**Deadweight loss** = ½ × tax × cups lost = ½ × 0.90 × 40 = 18 euros a day.

# Twice the tax, four times the loss

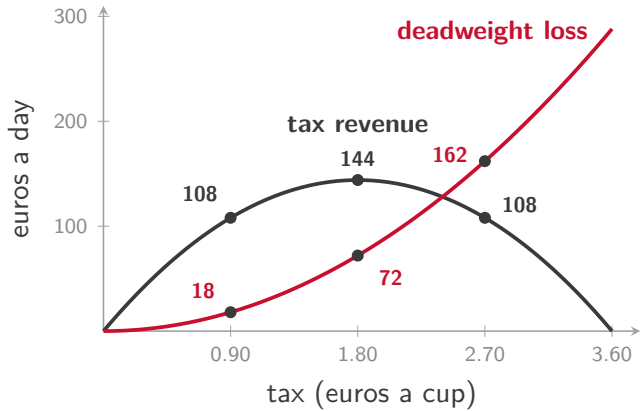


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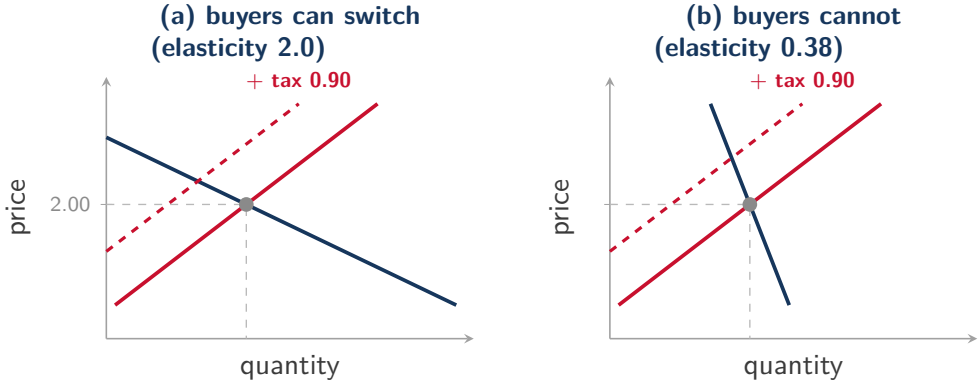

Twice the tax: twice the cups lost, twice the height — four times the loss.

# Raise the tax and revenue rises, then falls — the loss keeps growing

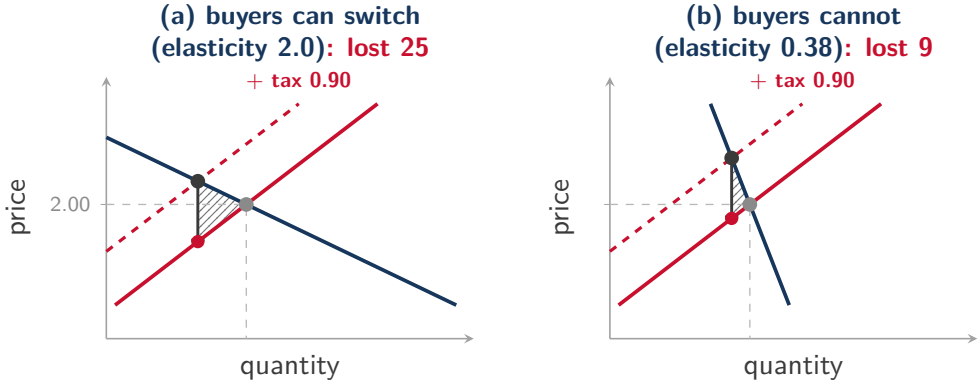



Beyond 1.80 a cup, a higher tax raises less money and loses more.

# The same 0.90 tax loses more where buyers can walk away



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The more elastic the market, the more units a tax stops — and the bigger the loss.

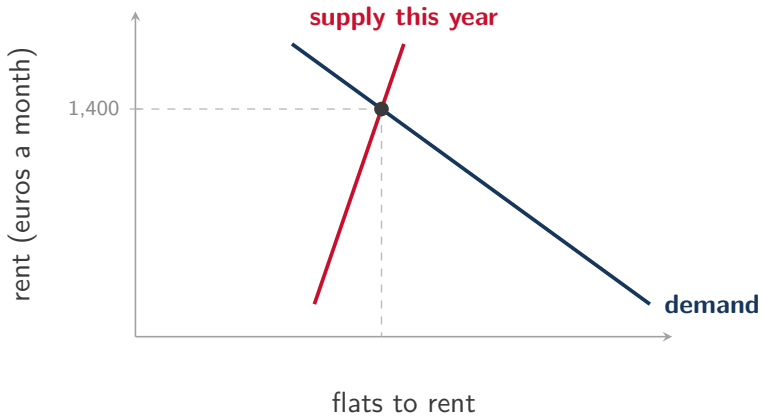
## In pairs, four minutes: what a tax on menús loses

### Same street, same menús

| menús a day | the last one is worth | the last one costs |
|---|---|---|
| 30 | 15.00 | 10.00 |
| 45 | 13.50 | 11.00 |
| 60 | 12.00 | 12.00 |

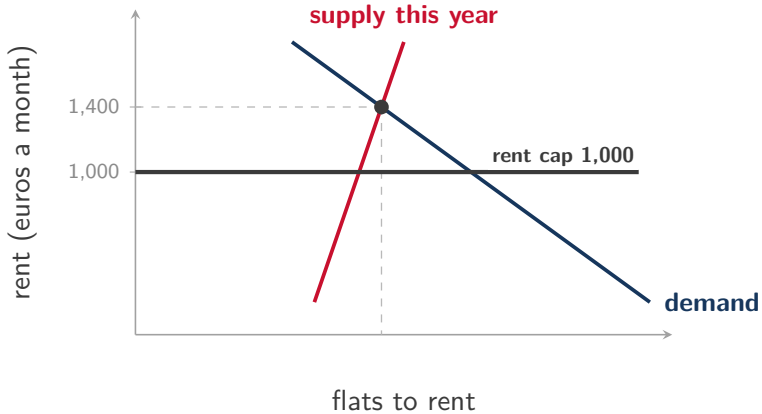
1. Tax 2.50: 45 menús. The lost surplus?
2. Tax 5.00: 30 menús. The lost surplus?
3. Revenue at each tax?

# A rent cap of 1,000 on Gràcia's street: 300 flats wanted, 200 offered



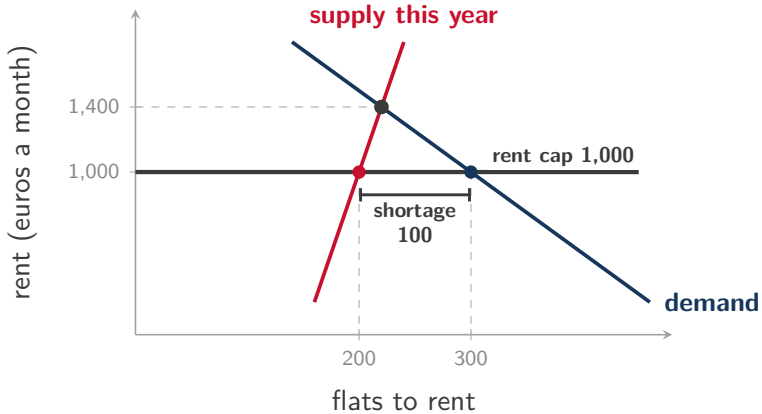

Own numbers (Class 7's Gràcia street); not data.

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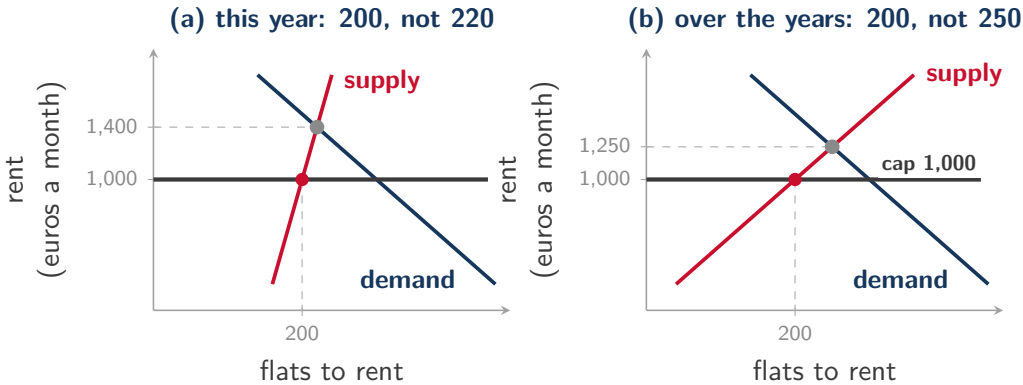
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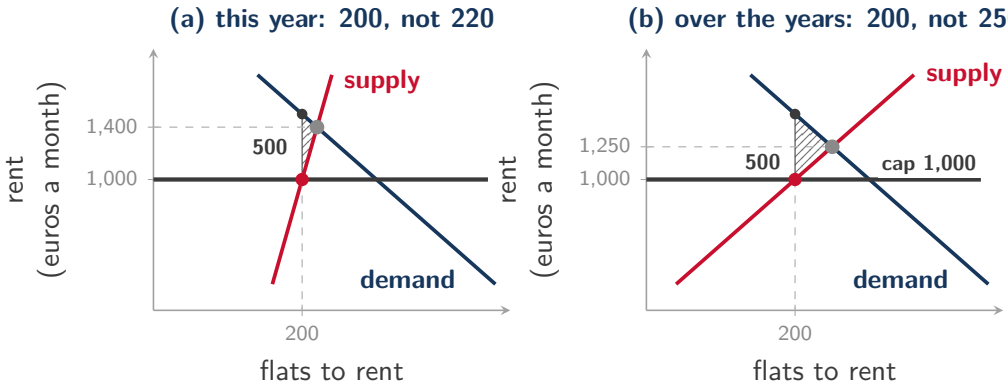

Below the market rent, a cap leaves 100 households searching.

Own numbers (Class 7's Gràcia street); not data.

# Over the years the cap costs 50 flats, not 20



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Lost surplus: ½ × 500 × 20 = 5,000 a month this year;
½ × 500 × 50 = 12,500 over the years.

# Revealed in class

Before class: from the cap picture, what should happen to rents on new contracts, and to the number of new contracts signed? Which would you expect to change first?

# Bottled water with a price cap (optional)

## Bottled water at a festival, price capped at 2 euros

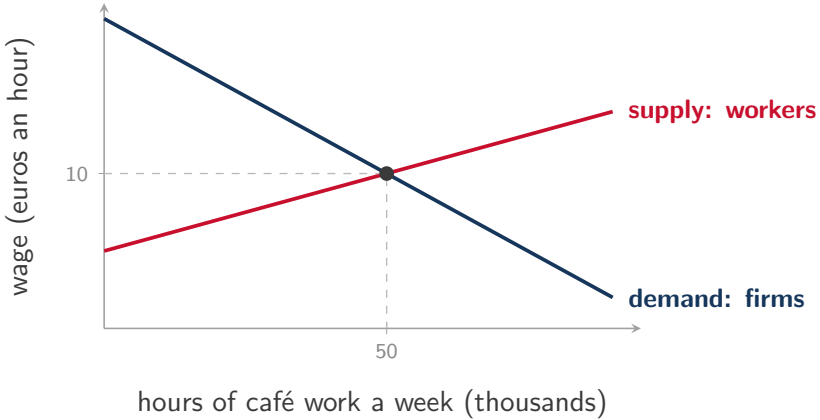
1. Market price 1.50: what does the cap do?
2. Heatwave: demand shifts right, the market price would be 3. Now?
3. Does the cap shift supply?

# Break

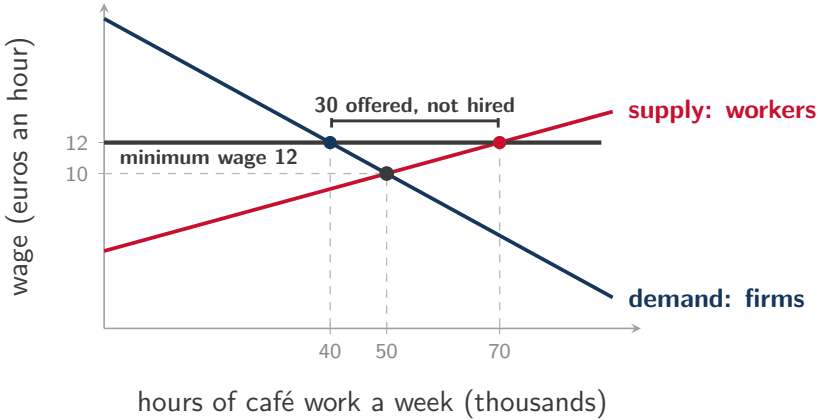
Ten minutes

Problem set 1: on the front desk as you go out

# A minimum wage is a price floor: workers offer 70, firms hire 40

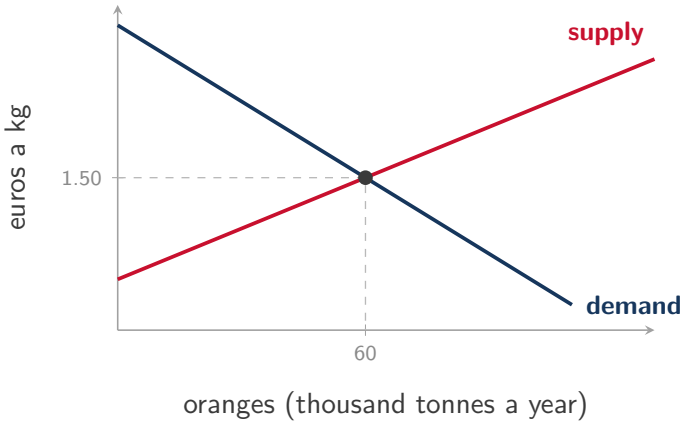


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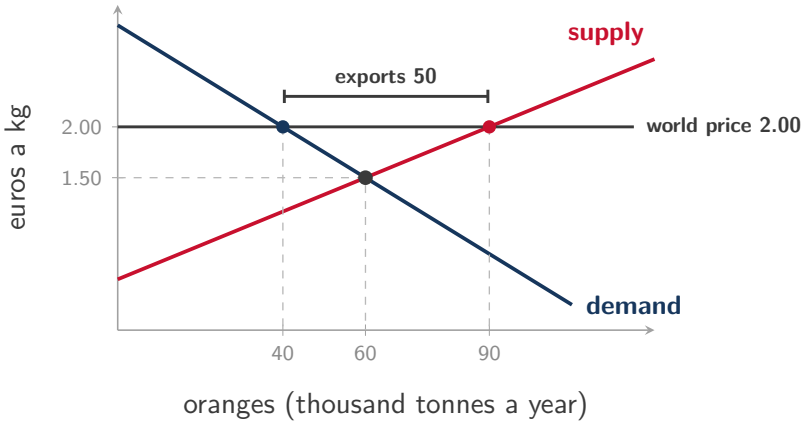



A floor above the market price: sellers offer more than buyers take — here, more hours than firms hire.

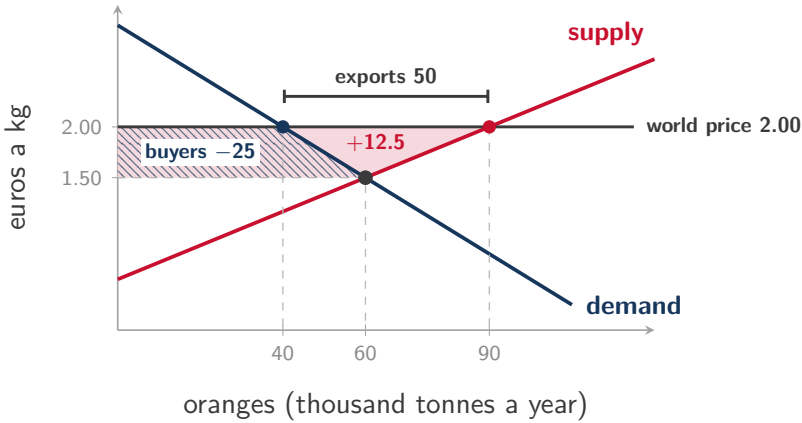
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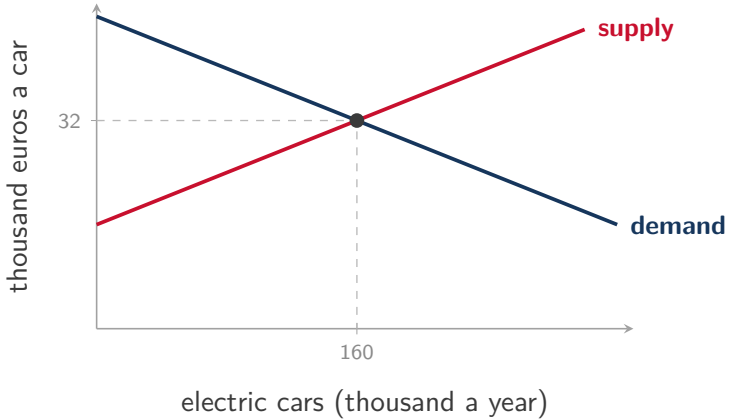
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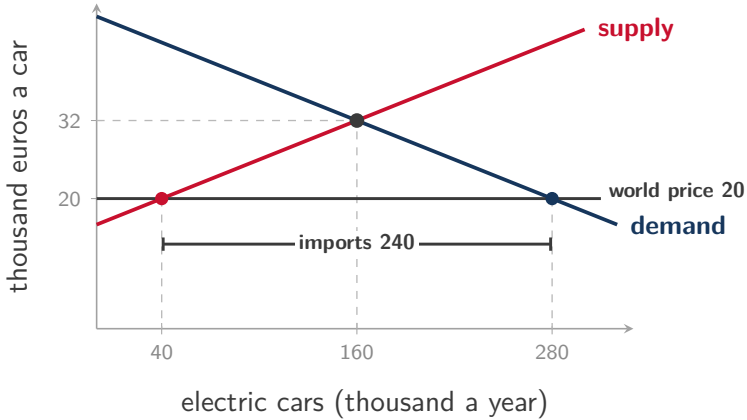

Sellers gain 37.5, buyers lose 25: the country gains 12.5 million euros a year.

1 thousand tonnes = 1 million kg, so euros a kg × thousand tonnes = million euros.

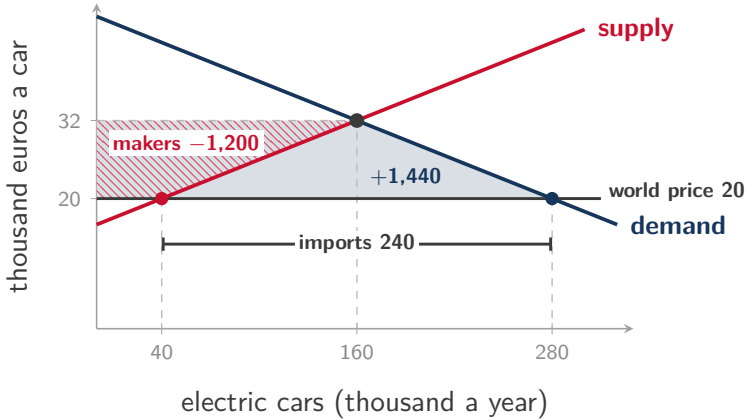
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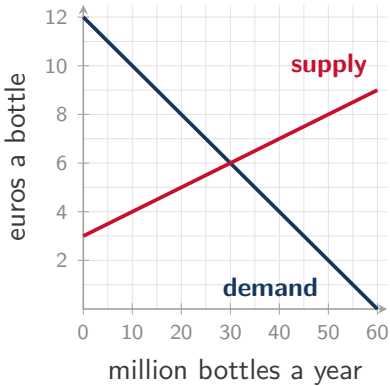
# World price below home price: imports; buyers gain, makers lose less




Buyers gain 2,640, home makers lose 1,200: the country gains 1,440 million euros a year.

1 thousand cars × 1 thousand euros a car = 1 million euros.

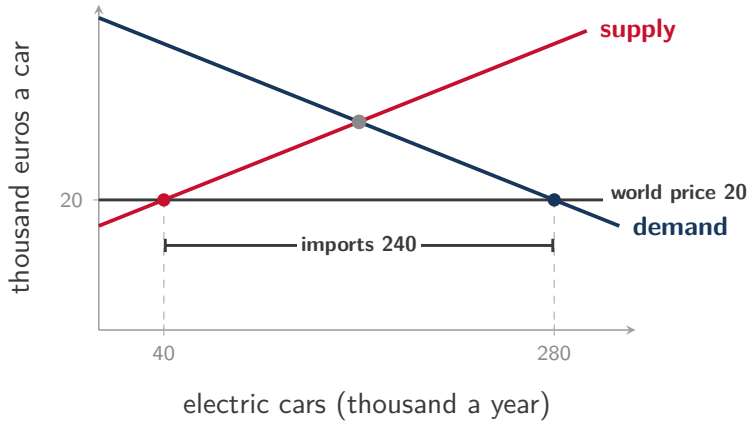
# In pairs, four minutes: cava at home and abroad



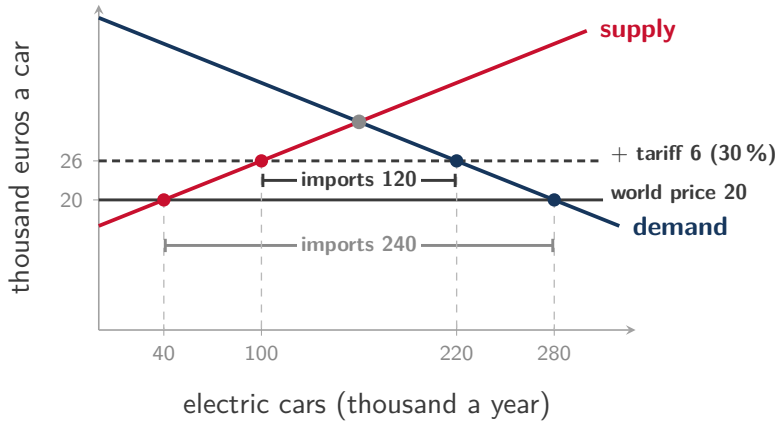

## Cava in a small country

1. No trade: price and bottles?
2. World price 8: export or import? How many?
3. At 8, who loses?
4. World price 4 instead?

# A tariff of 6 on imported cars: home price 26, imports halve

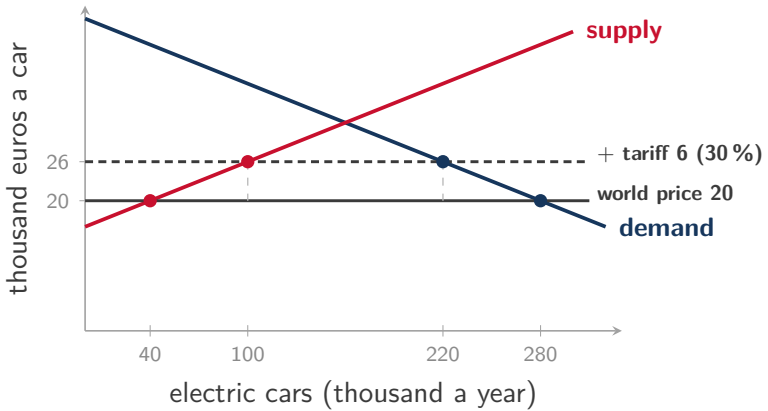


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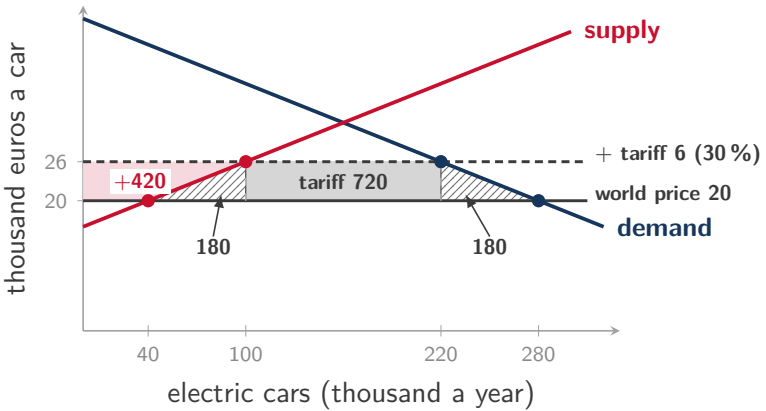



Home price 26: home makers 40 → 100, buyers 280 → 220, imports 240 → 120.

# The tariff's bill: 360 million a year that nobody gets



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Buyers lose 1,500 = makers 420 + state 720 + 360 that nobody gets (million euros a year).

# October 2024: the EU adds duties on electric cars made in China

| **EU's ordinary duty on cars (China included)** | **10 %** |
|---|---|
| **Plus, electric cars made in China:** | |
| BYD | 17.0 % |
| Geely | 18.8 % |
| SAIC | 35.3 % |
| other cooperating makers | 20.7 % |

February 2026: one China-made model exempted from the extra duty, for a minimum import price and a cap on numbers.

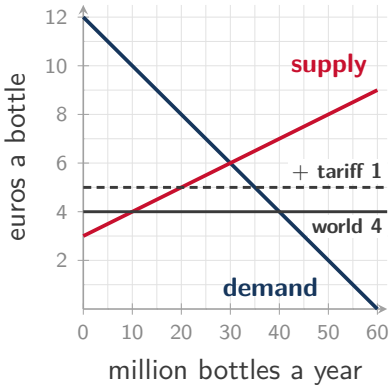
European Commission, IP/24/5589 (29 October 2024) and QANDA/24/3232 (12 June 2024); price undertaking accepted 10 February 2026.

## Five arguments for limiting trade — and the catch

| argument | the catch |
|---|---|
| **jobs** | export industries add jobs |
| **national security** | easily stretched to any industry |
| **infant industry** | hard to pick future winners |
| **unfair competition** | home buyers gain from cheap imports |
| **bargaining chip** | threats may have to be carried out |

Each protects someone; home buyers pay, two triangles are lost.

# In pairs, four minutes: a tariff of 1 euro on imported cava




## Cava: world price 4, tariff 1

1. Home price and imports?
2. Tariff revenue?
3. Surplus lost?
4. Who gains, and how much?

## Who exports steel? (optional)

| Wheat given up for one tonne of steel | |
|---|---|
| **North** | 4 tonnes |
| **South** | 2 tonnes |

North is faster at both goods.

1. Before trade, where is steel cheaper?
2. World price: 3 tonnes of wheat. Who exports steel?
3. Can North still gain from trade?

## Beyond the frontier (optional)

### An island: 10 hours of work a day

1 fish: 1 hour · 1 coconut: 2 hours · world: 1 coconut = 1 fish

1. It fishes 10 hours, swaps 3 fish for 3 coconuts. What does it eat?

2. Could it make that alone?

3. Inside, on or beyond its own frontier?

**A tax, a cap or a tariff shrinks a market; twice the wedge, four times the loss.**

**A binding cap: cheaper for some, a shortage for others.**

Open borders: the country gains, some lose; a tariff takes part of it back.

Monday: when a market hurts people outside it.

## Before Monday 26 October

- Monday: market failures, externalities and public goods
- Read Mankiw & Taylor: Ch. 9, pp. 204–216; Ch. 8, pp. 189–197 (pp. 197–199 optional)
- Problem set 2 (Block 3): handed out Monday; due Monday 2 November
- Practise in Sextant: the units the course page lists



pkocourek@ub.edu · pkocourek.com/teaching/intro-economics · pkocourek.com/sextant/?c=intro-econ